Rethinking the Arithmetic of Multi-Agent Medical Consultation

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Abstract

Multi-agent consultation is the default design pattern in medical language-model systems. Several agents are prompted into clinical specialties and deliberate until they agree, on the reasoning that makes a second opinion worth soliciting: when specialists who make different mistakes converge on a diagnosis, the convergence is itself evidence. Although that reasoning requires the specialists to make different mistakes, no evaluation of a medical agent panel has measured whether its members do. In this work, we measure it, across 480 multi-agent configurations and $161{,}999$ clinical episodes. They do not: a nine-member panel speaks with one voice on 66% of items and is wrong on 38% of what it agrees on. The failure decomposes into two components: a panel carries far less independent judgement than its size implies, and recovers almost none of the little it carries. Neither yields to its obvious remedy. Deliberation drives the agents to near-unanimity and lowers the accuracy of that unanimity. Crossing the vendor boundary, to six independently trained ecosystems on two continents, does not decorrelate their errors. And no aggregation rule we test beats counting votes, including one trained on the panel's own output against ground truth. In simpler terms: a medical panel that agrees is not a panel that knows, and its agreement must not reach a clinician as confidence.

1 Introduction

Medicine has a procedure for a hard case: convene several specialists, let each form a view, and see whether the views agree. What makes the procedure worth its cost is not that more people looked. It is that clinicians trained differently make different mistakes, so when they land on the same diagnosis the coincidence is unlikely unless the diagnosis is right. Concordance is evidence. That is why a tumour board's agreement moves a treatment decision, and why a second opinion matching the first ends the question rather than merely repeating it.

Multi-agent consultation transplants the procedure into language-model systems. MedAgents (Tang et al., 2024), MDAgents (Kim et al., 2024), MAC (Chen et al., 2025) and TeamMedAgents (Mishra et al., 2026) build the same object: several agents prompted into clinical specialties, deliberating until they agree, on the reasoning that a panel of specialists diagnoses better than one physician. The reasoning is an inheritance rather than an analogy, and inheritances carry conditions. This one requires the specialists to make different mistakes: pooling correlated judgements repeats them rather than combining them (Kurvers et al., 2016; Kämmer et al., 2017). Whether agents prompted into different specialties satisfy that condition has not been measured. The pattern is being deployed faster than its premise is being checked (Fig. 1).

In this work we measure it. A consultation is worth convening only if two things are true: somebody in the room is right when the others are wrong, and the room can tell who. The two are separately measurable, and collaboration gain is exactly their product. We measure each across 420 multi-agent configurations and $161{,}999$ episodes: eight models from four vendors, three medical benchmarks, five architectures, panel sizes $N \in \{1, 3, 5, 7, 9\}$, and a budget-matched single-model control at every cell. Medicine is what makes the measurement clean. Multiple-choice clinical reasoning calls no tools, so the whole difference between an architecture and its baseline on the same item is coordination—none of it the tool-call overhead that confounds the same comparison on general-domain agentic benchmarks. Scaling principles for general-domain agent systems (Kim et al., 2026) supply the architecture taxonomy and the controlled-comparison design; the error-correlation and consensus-calibration mea-

[FIGURE TO BE DRAWN]

What the paper shows, in one picture. Three panels, left to right, one clinical item running through all of them.

(a) What a consultation is supposed to be. Nine specialists, drawn as nine distinct figures, each reaching a view on their own. Two or three of them disagree with the majority, and one of the dissenters is holding the correct answer. Caption the panel with what makes this worth convening: *they trained differently, so they fail differently*.

(b) What an agent panel is. The same nine seats, but every figure is the same figure wearing a different label—one model, nine coats. Their views are near-identical. Annotate with $\varphi = 0.76$, $N_{\text{eff}} = 1.22$, and after deliberation $\varphi = 0.983$, $N_{\text{eff}} = 1.01$. The visual point is that the room *looks* full and is not.

(c) What reaches the clinician. One output box: a unanimous, high-confidence recommendation, rendered exactly the way a real MDT summary would be. Beside it, the two numbers that the box does not carry: unanimity 97.8%, wrong on 45% of what it agrees on. A clinician reading (c) cannot tell (a) from (b), and that is the finding.

Space: full text width, about 5 cm tall.

Figure 1: **A panel that agrees is not a panel that knows.** Nine agents prompted into clinical specialties carry 1.22 independent opinions between them, and 1.01 once they have deliberated. The consensus they emit is the output a clinician is most likely to act on, and it is wrong on 45% of the items it agrees on.

surements that carry our argument are new here.

Agents prompted into different medical specialties fail on the same items. Their pairwise error correlation is $\varphi = 0.76$, which leaves a nine-member panel carrying 1.27 effective independent opinions, and peer discussion raises the correlation to 0.986, at which point a nine-agent consultation is informationally equivalent to 1.01 physicians. The panel does not converge on an answer; it collapses onto one. Nor is the independence for sale. Crossing training families inside one vendor lifts a nine-member panel from 1.27 to 2.60 effective opinions, and crossing the vendor boundary altogether—to six independently trained ecosystems on two continents—returns 2.29. Two models built by different companies on different corpora fail on the same medical items about as reliably as two checkpoints from one lab.

The second condition fails harder, and it is the one prior work would not predict. Independence is scarce but not absent: at nine agents some member holds the correct answer 63.9% of the time against 53.1% for one doctor working alone, so 10.7 pp of real signal is in the room. Almost none of it comes out. Put the mechanisms on a scale where picking a panel member at random scores 0 and always picking a correct one scores 100. The five we test score between -12 and $+5$, and the two commonest—majority vote and confidence-gated referral—fall below zero. A classifier trained on the panel's own output against ground truth does not beat counting votes. What such a system would need is a signal separating a right member from a wrong one, and nothing the panel emits is one: stated confidence separates them at Youden $J = 0.166$.

For medicine the consequence is not inefficiency but hazard. A unanimous, high-confidence multidisciplinary recommendation is the output a clinician is most likely to act on, and it is the output whose reliability is least legible from its own signals. Unanimity separates the panel's right answers from its wrong ones at AUC $= 0.59$, and deliberation drives unanimity up while driving the accuracy of that unanimity down. The panel supplies the surface a clinician is trained to read as corroboration, and nothing behind it. Making medical panels agree is easy; making them *decidable* is the open problem.

The rest of the paper is organised as follows. Section 2 places the work against the collective-intelligence results the design pattern borrows from and the medical systems built on them. Section 3.1 defines the two measurements and the six conditions. Section 4.3 asks how much independent judgement a panel contains and prices every remedy for its absence; Section 4.4 asks what any aggregation rule recovers from it; and Section 4.5 reports what the resulting output does to the signals a clinician would act on.

2 Related work

Why pooled judgements help, and what they require. Collective-intelligence results in medicine rest on error independence: pooling independent diagnostic judgements improves accuracy (Kurvers et al., 2016; Wolf et al., 2015), plateauing at roughly five raters (Kämmer et al., 2017), and groups of nine outperform individuals on the Human Dx platform (Barnett et al., 2019), the clinical instance of a general principle (Surowiecki, 2004) that multidisciplinary tumour boards institutionalise (Radcliffe et al., 2019). The framing question is not whether two heads beat one but when (Hautz and Kämmer, 2024). The same literature documents what happens when independence fails: conformity to a unanimous majority (Asch, 1956) and groupthink in deliberating groups (Janis, 1972). Correlated errors across models bound ensemble benefit in general (Kim et al., 2025), and multi-agent debate can converge on confident-but-wrong answers (Zhu et al., 2026). This paper measures that failure directly in clinical panels, where diagnostic error is already common enough in ordinary practice (Graber, 2013; Singh et al., 2014) that manufactured agreement is a safety concern rather than an efficiency one. The strongest evidence that the independence premise is what matters comes from the hybrid case: pooling 40,762 physician differentials with five frontier models over 2,133 open-ended vignettes, Zöller et al. (2025) find that human–model collectives beat physician collectives *and* model ensembles, and attribute the margin to humans and models making different kinds of errors. Their result is the consequence; the quantity that produces it is the one we measure. It also establishes the regime: independence arrives when the added judgement comes from outside the model ecosystem, which is the boundary our diversity ladder (§4.3) locates from the inside.

Medical language models and multi-agent systems. Medical question answering became a capability benchmark once models matched licensing-exam thresholds (Kung et al., 2023; Nori et al., 2023; Singhal et al., 2023, 2025), and later work moved to conducting encounters (Tu et al., 2025; McDuff et al., 2025). Prompting strategy carries much of that progress (Nori et al., 2024), which is why architectural claims need budget-matched controls. On the multi-agent side, MedAgents (Tang et al., 2024) introduced role-played specialist collaboration, MDAgents (Kim et al., 2024) added adaptive routing, and MAC (Chen et al., 2025) and TeamMedAgents (Mishra et al., 2026) formalised teamwork components, the latter sweeping team sizes 2–5. Every published medical ablation stops at $N \leq 5$, none reports a dose–response curve, and none runs a budget-matched single-model control. Skeptical evaluations find that orchestration often fails to beat well-prompted single models at substantially higher cost (Zhu et al., 2025; Liu et al., 2026; Zhao et al., 2025); our independence measurement supplies the mechanism those results lack. The pattern has since diversified along every axis except the one measured here. Chen et al. (2024b) and Zhou et al. (2025) specialise the roster—an MDT for rare disease, role-assigned radiologist and generalist agents for multimodal input—and Liu et al. (2025) adds a director to coordinate them. Ma et al. (2025) replaces majority voting with logical contradiction resolution and Li et al. (2025) lets agents accumulate diagnostic knowledge across a cohort, both of which are attempts to escape a voting rule rather than to escape the correlation that makes voting weak. Fan et al. (2024), Sanghvi et al. (2026) and Sviridov et al. (2025) move the setting itself, from single-shot selection to simulated, interactive and multimodal encounters where agents must ask before they answer, and Chen et al. (2026) and Shi et al. (2026) carry the pattern into inpatient pathways and outcome prediction, where triage, diagnosis and treatment are separate coordinated stages. These are complementary: each varies the aggregation rule, the roster or the interaction protocol, and each is a candidate for raising the capture rate we define in §4.4. None of them measures the error correlation among its own agents, so none can say whether the signal its mechanism is competing for is present.

Closest to our central question, Yuan et al. (2026) compare single-vendor and mixed-vendor panels of three frontier models on rare-disease diagnosis and report that mixed-vendor teams win, attributing the gain to complementary inductive biases surfaced by an overlap analysis. That is the same lever we isolate in Table 4, measured with a different instrument and in the regime our results predict is the favourable one—three models of comparable strength, where decorrelation carries no quality penalty. Two of our findings bear on how such comparisons should be read. Raw overlap and raw correlation are depressed mechanically when members differ in accuracy (Eq. 2), so vendor-diversity effects should be reported normalised; and the lever is small wherever we can measure it, moving a nine-member panel from 1.27 to 2.60 effective opinions within one vendor and to 2.29—not further, and in fact back—once the panel spans six ecosystems on two continents.

Scaling agents, and scaling inference. Debate protocols improve factuality (Du et al., 2024; Liang et al., 2024) and frameworks such as MetaGPT (Hong et al., 2024) and AutoGen (Wu et al., 2024) made the pattern easy to deploy (Zhang et al., 2024). On how far it scales, "more agents is all you need" (Li et al., 2024) reported monotone gains, while Chen et al. (2024a) showed the relationship is non-monotone and Li et al. (2026) confirmed an inverted-U in agent count; Yang et al. (2026) attributed the shape to opinion diversity, showing two diverse agents can match sixteen homogeneous ones. Kim et al. (2026) fit a predictive scaling principle across architectures, model families and agentic benchmarks, and supply the architecture taxonomy and controlled-comparison design our grid is built on. Test-time compute offers the alternative to adding agents (Kaplan et al., 2020; Hoffmann et al., 2022; Brown et al., 2024; Snell et al., 2024), and self-consistency (Wang et al., 2023) over chain-of-thought (Wei et al., 2022) is the budget-matched control we run at every cell.

Confidence and calibration. Our safety analysis depends on whether stated confidence is informative. Models are partially aware of what they know (Kadavath et al., 2022), but elicited confidence is weakly calibrated and sensitive to how it is asked (Xiong et al., 2024; Guo et al., 2017), and clinical evaluations report fluent but unreliable responses (Goodman et al., 2023) with demographic biases that propagate (Omiye et al., 2023; Zack et al., 2024). We add a coordination-specific finding: deliberation degrades the discriminative power of confidence rather than improving it.

3 Measuring what a consultation adds

3.1 The measurement problem

A panel that pools complementary evidence and a panel that repeats itself produce the same observable: several agents, an aggregation step, one answer. Accuracy cannot separate them. A panel can gain a point because its members caught each other's errors, or because nine correlated draws from one distribution happened to land on the mode more often than one draw did, and the two are indistinguishable from the output alone. This is why the multi-agent medical literature can report gains and the sceptical literature can report that orchestration fails to beat a well-prompted single model (Zhu et al., 2025; Liu et al., 2026; Zhao et al., 2025) without either side being wrong. The disagreement is about a quantity neither measures.

The premise the design pattern inherits is specific enough to test. Pooling judgements improves accuracy when the judgements are independent; correlated judgements are merely repeated. So we instrument each episode with the two quantities that premise depends on, and report them alongside accuracy throughout.

How many opinions a panel actually carries. Writing φ for the mean pairwise error correlation between panel positions—the correlation of their per-item error indicators—the effective number of independent opinions in a panel of N agents is

$$N_{\text{eff}} = \frac{N}{1 + (N-1)\varphi}. \tag{1}$$

At $\varphi = 0$ the N opinions are independent and $N_{\text{eff}} = N$; at $\varphi = 1$ they are one opinion copied N times and $N_{\text{eff}} = 1$. This is the standard design-effect correction for clustered observations, and applying it to a panel of agents replaces an assumption with a measurement: a system that reports $N = 9$ is making a claim about how much judgement is in the room, and Eq. 1 is what that claim is worth.

Raw φ has a trap that matters for any comparison of diverse panels. Two binary error indicators with unequal marginals cannot reach $\varphi = 1$: with error rates $p \leq q$ the attainable maximum is

$$\varphi_{\max} = \sqrt{\frac{p\,(1-q)}{q\,(1-p)}}, \tag{2}$$

which falls as the members' accuracies diverge. A panel assembled from models of different strength therefore posts a lower raw φ whether or not its members reason differently. We report $\varphi/\varphi_{\max}$ wherever members differ in accuracy.

What a panel holds, and what its aggregation keeps. Independence sets what is available; an architecture decides what is delivered. Let the *oracle ceiling* P_{oracle} be the accuracy of a rule that selects a correct first-round opinion whenever any member holds one—an upper bound on every aggregation rule that cannot see the label. Collaboration gain then factors exactly:

$$\underbrace{P - P_{\text{SA}}}_{\text{gain}} = \underbrace{(P_{\text{oracle}} - P_{\text{SA}})}_{\text{headroom}} \times \underbrace{\kappa}_{\text{capture}}, \tag{3}$$

where P_{SA} is the single-doctor baseline on the same items and κ is the fraction of the headroom an architecture recovers. On this scale $\kappa = 0$ is the accuracy of a panel member drawn at random and $\kappa = 100$ is the oracle, so a negative κ means the aggregation rule is worse than picking a member and believing them. The factorisation is definitional. Its use is that the two factors have different causes and different remedies: headroom is set by how much independent judgement the panel contains, capture by whether anything can arbitrate between members who disagree. A system that fails on one is not helped by fixing the other.

We also record *error amplification* A_e: how often the panel held the correct option among its first opinions and emitted a wrong one anyway, normalised by the marginal probability that a single agent is wrong. A_e is κ seen per episode rather than per configuration, and it identifies the failure as destruction of information the system already possessed rather than absence of it.

3.2 Conditions

An agent system is a triple (A, C, Ω): agents, a communication graph, and an aggregation rule (Fig. 2). The taxonomy is the one used for general-domain agent systems (Kim et al., 2026); the clinical instantiations are ours. Every condition below shares the item rendering, the answer schema, the role prompts and the decoding parameters, so only the coordination logic varies. Each is stated with what it instantiates clinically and what it is for in the experiment.

SINGLE. One agent answers, zero-shot or with chain-of-thought. The baseline P_{SA} against which every gain in this paper is measured, run on the same items in the same cell as the panels it is compared to.

SELF-CONSISTENCY. The same model, the same *generic* prompt, k samples at the same temperature, aggregated by majority vote at a matched dollar budget. This is the control that makes the roster identifiable. It produces k answers from one model with no specialty prompts at all, so comparing it to a k-member panel isolates what the specialty roster contributes over drawing the same model k times.

INDEPENDENT. N specialists receive the item and answer without seeing one another; aggregation is majority vote, ties broken by mean stated confidence and then option order. The pooling arithmetic in its bare form. With $C = \emptyset$ the roster is the only thing that can make two panelists disagree, which makes this the arm where role prompting is measured without a channel confounding it.

CENTRALIZED. An orchestrator receives the N opinions on a star topology and must *audit before aggregating*: name the single opinion most likely to be wrong and justify that, then either commit or re-task one named specialty, which re-answers with the critique in hand. Specialists never see each other, so the star is preserved. This gives the panel one designated point at which an error can be caught before it reaches the output—the attending physician—and tests whether a designated auditor is what capture requires.

DECENTRALIZED. N specialists give first opinions, then exchange them on a complete graph for up to three rounds, each free to revise. Rounds stop early only on true unanimity, defined as every agent emitting the same *valid* option; a panel containing an unparsed answer is not unanimous. This is the multidisciplinary team meeting, and the arm in which independence and agreement can be measured before and after the same agents talk to each other.

HYBRID. A generalist answers first with a stated confidence; below a frozen threshold θ the case is referred to the N-specialist panel, and a panel with no majority escalates to full discussion. θ is calibrated once on a 100-item development slice disjoint from every evaluation set. This is tiered referral, and it is the only condition that decides *whether* to convene a panel, which makes it a direct test of whether a system can route on its own confidence.

GENERIC PANEL. INDEPENDENT, CENTRALIZED and DECENTRALIZED rebuilt out of undifferentiated internists: every member receives the same generic system prompt, so the specialty roster is the single removed ingredient and panel size is the only thing that varies. This separates what a panel gains from being several agents from what it gains from those agents being told they are different specialists.

Panel composition and nesting. A cheap router (`gpt-4.1-nano`, temperature 0) ranks the nine most relevant specialties for each item from a fixed 17-specialty roster; a panel of size N is the first N of that ranking. Every agent's first opinion is

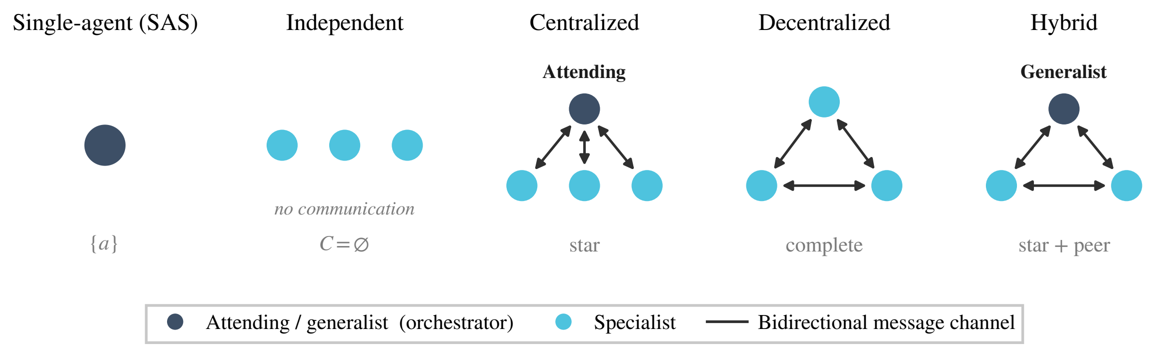

Figure 2: **The coordination structures we evaluate**, with the clinical role each instantiates. A is the agent set, C the communication graph, Ω the aggregation rule. Two further conditions vary no topology and so do not appear here: SELF-CONSISTENCY draws the same model k times with no specialty prompt, and GENERIC PANEL reruns the three middle structures with the roster removed.

keyed on $(\text{item}, \text{specialty}_i, \text{seed}, i)$ and never on N or on the architecture, so panels are *nested*—the $N{=}9$ panel contains the $N{=}5$ panel—and all four multi-agent conditions reuse the same first opinions. This is a deliberate common-random-numbers coupling: it removes between-panel sampling noise from every architecture and panel-size contrast, and it fixes the estimand as *accuracy as one routed panel is extended* rather than the expected accuracy of independently redrawn panels. Paired McNemar tests remain valid under the coupling; regressions cluster at the item level.

3.3 Clinical tasks

Three benchmarks, 250 items each, sampled with a fixed seed: MedXpertQA-Text (Zuo et al., 2025) (10 options; stratified by the 12 body systems × 3 task types), MedQA-USMLE (Jin et al., 2021) (4 options), and the contamination-screened hard split of MedAgentsBench (Tang et al., 2025) (894 items excluding the `medqa_5options` subset; the first 250 after stratified shuffling). The endpoint is exact match against the dataset's own gold key. No LLM judge appears anywhere in the evaluation path.

Multiple-choice clinical reasoning is what makes the measurement clean. Agent systems on general-domain benchmarks call tools, and tool-call overhead dominates the token accounting and confounds any comparison of coordination structures; a panel that coordinates more also retrieves more, and the two effects are not separable from the output. Here no condition calls a tool, so the entire difference between an architecture and the single-doctor baseline on the same item is coordination. The cost is external validity, which we state in §6: a real consultation involves examination, imaging and the option of ordering another test, and this setting has none of them. What it does have is the property that the arithmetic premise is measurable at all.

Every item is additionally tagged easy / medium / hard by the pass rate of three mid-tier models at $k{=}5$ samples. On MedXpertQA the strata are heavily skewed hard (13 / 62 / 125 on the pilot sample), which is the intended regime.

4 Experiments and results

4.1 Setup

LLMs and capability scaling. Three OpenAI models form the capability ladder the architecture grid is run on, measured rather than assumed (Table 8; the screening probe over eight candidates is Table 10). The ladder is monotone on both benchmarks and spans single-doctor accuracy from 25% to 55% on MedXpertQA, straddling the previously reported 45% threshold from both sides. Five further models—Gemini 3.5 Flash Lite and Gemini 3.7 Flash from Google, Claude Haiku 4.5 and Claude Sonnet 5 from Anthropic, and `deepseek-v4-flash` from DeepSeek—are run on the same items and the same 27-cell grid. Five more—`deepseek-v4-pro`, `qwen3.8-flash` and `qwen3.8-max`, `glm-5.3-flash` and `glm-5.3`—contribute single-doctor baselines on the same items, which is what the diversity analysis of §4.3 consumes. Thirteen models across six independently trained ecosystems span $I = 34.0$ to 70.9, and the ladder pairs an American and a Chinese model at matched capability at several rungs—`deepseek-v4-pro` at 59.7 beside `gpt-5-mini` at 59.2, Gemini 3.5 Flash Lite at 50.3 beside `gpt-5-nano` at 50.8—so the vendor contrast can be drawn at matched capability rather than confounded with it. A task-grounded capability metric explains more variance than a generic intelligence index (Kim et al., 2026), so we take I to be each model's mean single-doctor chain-of-thought accuracy across the three benchmarks.

Budget-matched controls. Every configuration is accompanied by three single-model controls. These are zero-shot, chain-of-thought, and **self-consistency at a matched budget**. The self-consistency pool is built from cached $n{=}8$ sampling calls (the API maximum), so any k is a prefix of the same samples. Nominal cost is charged as $\lceil k/8 \rceil$ prompts plus k completions, which is what an efficient implementation would actually pay. Panels are compared against the self-consistency cell whose mean per-item *dollar* cost is closest, not the one with a matching sample count. Matching on samples silently favours the panel, because a panel pays the prompt once per agent while self-consistency amortises it across a batch.

Cost accounting is *nominal* throughout. Cached calls are charged at list price, so the economics figures report what a configuration would cost to run, not what our cache made it cost.

Statistics. Paired McNemar tests with exact binomial p-values on identical item sets, Holm-corrected within each family of comparisons. Wilson 95% confidence intervals for all accuracies. Power was calibrated on a 200-item pilot. Median discordance between configurations is 8.5%, giving a minimum detectable difference of 5.7 percentage points at $n{=}200$ and 2.6 points at $n{=}1000$ with 80% power. Configurations that fail for infrastructure reasons (rate limits, timeouts) are recorded with an error status and excluded from accuracy. Scoring them as wrong answers would bias call-heavy architectures downward.

4.2 Main results: what a panel buys

Across 480 multi-agent configurations and 161,999 episodes, convening a panel changes the answer on a minority of cells and never by much. Figure 3 gives the full dose–response surface; Table 1 the underlying numbers.

Which panel you convene does not matter. Taking each architecture at its best panel size in each of the 21 cells gives 126 pairwise architecture contrasts, of which **five are significant** (paired McNemar $p < 0.05$, largest difference 5.6 pp). Against the single doctor, 12 of 84 contrasts reach significance. The topology—star, complete, tiered, or no edges—does not separate at all, and the ordering does not carry across benchmarks: Kendall's τ between the architecture rankings of any two benchmarks averages $+0.067$, and leave-one-benchmark-out cross-validation gives $R^2 = -0.015$. An architecture chosen on one clinical benchmark carries no guarantee on another. Panel size behaves the same way. On `gpt-5-nano` / MedXpertQA, Independent moves from 31.8% at $N{=}1$ to 32.6% at $N{=}9$, which is 0.8 pp for nine times the calls, while Decentralized reaches 38.0% at $N{=}3$ and stays there. Gains saturate at $N{=}3$ in every cell where they occur, and no architecture in any cell improves beyond $N{=}5$ (Fig. 10, Fig. 11).

Both factors are small, and they move apart. A consultation is worth convening only if somebody in the room is right when the others are wrong, and the room can tell who. Eq. 3 separates those two conditions (Fig. 4), and neither holds. What is not definitional about the factorisation is that the two factors move in opposite directions (Table 2): headroom is largest where the single doctor is weakest, from $+14.5$ pp at $P_{\mathrm{SA}} \in [25, 50)$ down to $+4.1$ above 70%, while capture runs the other way, from -9.2% to $+24.7\%$. The panels with the most to recover are the ones that discard it, and the product stays between -0.3 and $+1.2$ pp across every band.

The rest of this section measures the two factors in turn. §4.3 asks how much independent judgement a panel of role-prompted agents actually contains, which is what sets the headroom, and prices every remedy for its absence. §4.4 asks what any aggregation rule recovers from it. §4.5 reports what the resulting output does to the signals a clinician would act on.

4.3 Is anyone in the room right? The arithmetic premise

Majority voting converts independent judgements into accuracy; correlated judgements it merely repeats. The headroom in Eq. 3 is therefore a measurement of how much independent judgement a panel contains.

Somebody in the room is right, and it is not the specialists. At $n_a{=}9$ the single doctor scores 53.1% and the oracle ceiling 63.9%: the panel

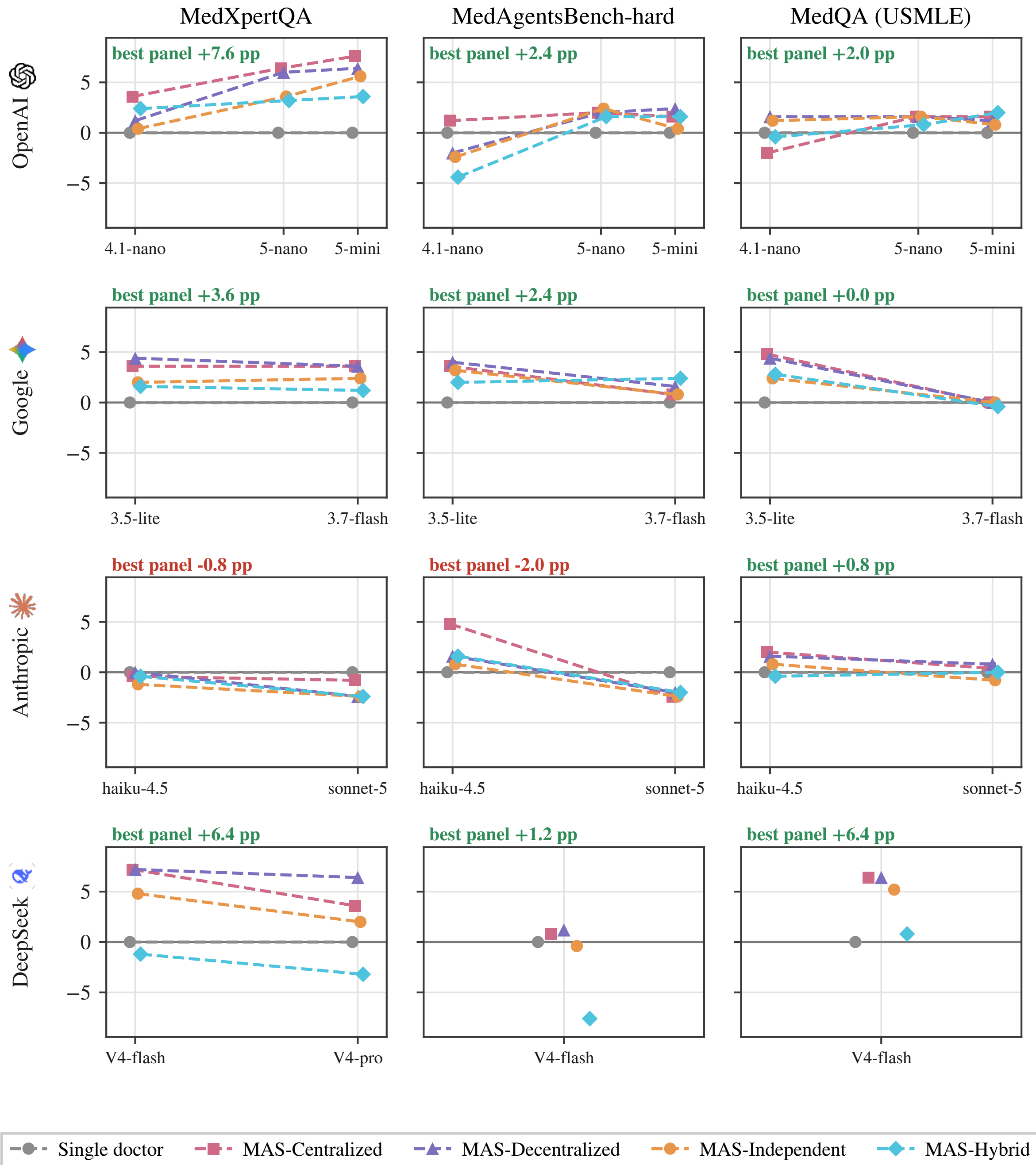

Figure 3: **Agent scaling across model capability and system architecture.** Gain over the single doctor, in percentage points, against the task-grounded capability index I (mean single-doctor accuracy across the three benchmarks); each architecture at its best-performing panel size. Rows are vendors, columns benchmarks, and all twelve panels share one vertical scale, so magnitudes are comparable throughout. Plotting absolute accuracy instead would put the three benchmarks on ranges of 15–95% and collapse the architectures onto a single line: the effects being compared span -4.4 to $+7.6$ pp, while capability moves performance by tens of points. Every configuration sits inside a narrow band around the baseline, and the band narrows as the single doctor improves.

Table 1: **Main results.** For each model and benchmark: the single-doctor baseline, then each architecture at its best panel size, and the gain of the best architecture over the single doctor. Bold marks the best architecture in each row. The complete 480-configuration grid, including every panel size, is released with the code (§E).

Model	Single doctor	Best multi-agent variant (accuracy % / n_a) Independ.	Central.	Decentr.	Hybrid	Gain best
MedXpertQA — expert-level reasoning, 10 options						
gpt-4.1-nano	15.6	16.0 / 9	**19.2** / 1	16.8 / 5	18.0 / 9	+3.6
gpt-5-nano	32.0	35.6 / 5	**38.4** / 3	38.0 / 7	35.2 / 5	**+6.4**
gpt-5-mini	41.6	47.2 / 5	**49.2** / 3	48.0 / 5	45.2 / 1	**+7.6**
gemini-3.5-flash-lite	34.8	36.8 / 5	38.4 / 5	**39.2** / 7	36.4 / 9	+4.4
gemini-3.7-flash	60.4	62.8 / 3	**64.0** / 3	**64.0** / 3	61.6 / 3	+3.6
claude-haiku-4.5	23.2	22.0 / 9	22.8 / 1	**23.2** / 7	22.8 / 1	+0.0
claude-sonnet-5	49.6	47.2 / 3	**48.8** / 1	47.2 / 5	47.2 / 1	-0.8
deepseek-v4-flash	37.2	42.0 / 9	**44.4** / 3	**44.4** / 5	36.0 / 3	**+7.2**
MedAgentsBench-hard — items where strong models still fail						
gpt-4.1-nano	20.4	18.0 / 1	**21.6** / 7	18.4 / 5	16.0 / 9	+1.2
gpt-5-nano	32.4	**34.8** / 5	34.4 / 3	34.4 / 9	34.0 / 5	+2.4
gpt-5-mini	44.4	44.8 / 9	46.0 / 7	**46.8** / 9	46.0 / 1	+2.4
gemini-3.5-flash-lite	31.2	34.4 / 3	34.8 / 3	**35.2** / 3	33.2 / 7	+4.0
gemini-3.7-flash	56.4	57.2 / 9	57.2 / 9	58.0 / 9	**58.8** / 9	+2.4
claude-haiku-4.5	20.8	21.6 / 9	**25.6** / 3	22.4 / 5	22.4 / 7	+4.8
claude-sonnet-5	47.6	45.2 / 9	45.2 / 7	**45.6** / 9	**45.6** / 7	-2.0
deepseek-v4-flash	39.6	39.2 / 9	40.4 / 7	**40.8** / 7	32.0 / 9	+1.2
MedQA (USMLE) — licensing-exam questions, 4 options						
gpt-4.1-nano	66.0	67.2 / 9	64.0 / 9	**67.6** / 5	65.6 / 9	+1.6
gpt-5-nano	88.0	**89.6** / 5	**89.6** / 5	**89.6** / 9	88.8 / 5	+1.6
gpt-5-mini	91.6	92.4 / 1	93.2 / 1	92.8 / 5	**93.6** / 1	+2.0
gemini-3.5-flash-lite	84.8	87.2 / 9	**89.6** / 3	89.2 / 7	87.6 / 9	+4.8
gemini-3.7-flash	96.4	**96.4** / 7	**96.4** / 3	**96.4** / 3	96.0 / 9	+0.0
claude-haiku-4.5	79.6	80.4 / 9	**81.6** / 5	81.2 / 3	79.2 / 9	+2.0
claude-sonnet-5	94.0	93.2 / 9	94.4 / 9	**94.8** / 5	94.0 / 5	+0.8
deepseek-v4-flash	88.0	93.2 / 7	**94.4** / 9	**94.4** / 7	88.8 / 7	**+6.4**

Table 2: **Gain decomposed.** Collaboration gain is headroom times capture rate (Eq. 3). The two factors move in opposite directions, so the product is small throughout. n is configurations.

P_{SA} band	n	headroom	capture κ	gain
$< 25\%$	56	+6.7 pp	−12.6%	−0.29
25–50%	144	+14.3 pp	−9.0%	+0.95
50–70%	44	+6.3 pp	+10.6%	+0.68
$> 70\%$	92	+4.1 pp	+24.7%	+1.20

holds 9.3 pp of recoverable accuracy, often enough to matter.

Where that headroom comes from is the surprise. Self-consistency gives one model the same generic prompt nine times at one temperature; a panel gives it nine different specialty prompts. Both produce nine answers, so both admit the same oracle measurement. Nine samples of one generalist put a correct answer on the table 63.5% of the time; nine distinct specialists put one there 63.9% (Fig. 5a). Of the 10.7 pp, **10.4**—97%—is already available from sampling one generalist repeatedly, and the specialty roster contributes +0.37 pp. The specialty roster is the defining feature of the systems built on this design pattern, and what it adds to the room is not measurable beside a temperature setting.

A nine-member panel carries 1.2 independent opinions. Across the three architectures that convene a full panel on every item at $n_a \geq 3$ — Independent, Centralized and Decentralized—the mean pairwise error correlation is $\varphi = 0.758$, and it is the same to three decimals in each of them separately. (Hybrid is excluded here and reported separately: its triage gate convenes a panel on only 21% of items, so its φ is measured on a harder subset.) By Eq. 1 this leaves $N_{\text{eff}} = 1.20$ at N=3, 1.25 at N=5 and 1.28 at N=9 (Table 3). Tripling the panel adds 0.08 of an independent opinion. Specialty role prompts do not make one model into several physicians; they make it into one physician wearing several labels.

[FIGURE TO BE DRAWN]

The two questions, and where each is answered. Three rooms side by side, nine seats each, with the panel's emitted answer in a box below each room.

Left — no headroom. Every seat holds the same wrong answer; the emitted answer is wrong. No aggregation rule can help. Label: *headroom* $= 0$.

Middle — headroom, no capture. Two seats hold the correct answer, seven do not, and the emitted answer is still wrong. Label: *headroom* > 0, $\kappa \approx 0$. Mark this room — it is the one the grid lands in.

Right — headroom and capture. The same two correct seats, and the emitted answer is correct. Label: *headroom* > 0, κ high.

Underneath, Eq. 3 spanning the three rooms, with *headroom* sitting under the seats and κ under the arrow from seats to output box. Point the reader at §4.3 beneath the seats and §4.4 beneath the arrow.

The takeaway before any number: the left and middle rooms are indistinguishable from the output alone, and separating them is what the rest of this section does.

Space: one column, about 4.5 cm tall.

Figure 4: **A panel can fail two ways, and accuracy does not tell them apart.** Either the answer is not in the room, or it is and nothing recovers it. The two have different remedies. §4.3 measures the first factor and §4.4 the second.

Table 3: **Effective independent opinions.** φ is the mean pairwise error correlation between panel positions; $N_{\text{eff}} = N/(1 + (N-1)\varphi)$. Neither growing the panel nor changing how it is wired moves φ. The shaded row is the single exception: Hybrid's triage gate declines to convene a panel on most items, so its measured pairs come from the harder subset it does escalate.

Architecture	N	φ	N_{eff}	info. used
Growing one panel: tripling N buys 0.08 of an opinion				
Independent	3	0.756	1.19	40%
Independent	5	0.758	1.24	25%
Independent	9	0.762	1.27	14%
Changing the architecture at N=9: no effect either				
Centralized	9	0.756	1.28	14%
Decentralized	9	0.756	1.28	14%
Hybrid	9	**0.595**	**1.56**	17%

Discussion drives the effective panel size to one. Peer discussion raises φ from 0.749 before the exchange to $\mathbf{0.986}$ after it (Fig. 6) (paired $t = 26.5$, $p = 2 \times 10^{-13}$). At $\varphi = 0.986$ a nine-agent panel has $N_{\text{eff}} = 1.01$. A multidisciplinary consultation, at the moment it reaches consensus, is informationally equivalent to one physician, and the vote taken at that moment is a vote among copies. This is the mechanism behind every safety result in §4.5: unanimity rises to 98% because the opinions have merged, not because they have converged on truth; the error rate given unanimity stays above 45% because merging does not add information; and no signal the panel emits marks the difference, because N agents agreeing looks like corroboration when it is repetition.

What discussion actually moves. The collapse is not agents falling silent. Across 18,196 (item, panel position) pairs, 38.6% of opinions change between the first round and the last, so deliberation is doing something. Where it takes them is the point. Of the changes, 34.6% go from wrong to right, 22.2% from right to wrong, and $\mathbf{43.2}$% from one wrong answer to a different wrong one. Two-thirds of all movement ends somewhere other than the correct option. The net effect on individual opinions is mildly positive—872 more corrections than corruptions, or +4.8 pp of positions—and the net effect on the panel is not, because the same exchange that corrects an individual is what makes the others agree with him. Deliberation buys a small improvement in each member and pays for it with the independence that made pooling members worth anything.

This is also why panel size predicts nothing. Over $N \in \{3, \ldots, 9\}$, N_{eff} moves from 1.20 to 1.28: the quantity that matters barely varies, so the quantity a designer controls cannot predict. Fitted against every configuration, panel size alone reaches cross-validated $R^2 = -0.006$, worse than predicting the mean (§F).

Role prompting is the panel's only diversity source until a channel replaces it. The oracle measurement asks what reaches the room; the complementary question is what a panel delivers once an architecture acts on it. We answer it by rebuilding the panel out of undifferentiated internists: every member receives the same generic system prompt, so panel size is the only thing that varies and the specialty roster is the single removed ingredient. Arms share the model, the temperature, the architectures and the same 250 items (`gpt-5-nano`, MedXpertQA), and we pair them over the 12 matched (architecture, $N \geq 3$) cells. Against one doctor at 32.0%, the specialist panel gains +4.63 pp. Asking an internist N times supplies +2.77 pp of that (95% CI [+1.51, +4.03], $p = 0.0005$) and the roster supplies +1.87 pp (95% CI [+0.95, +2.78], $p = 0.0009$)—so role prompting is worth just under two points of accu-

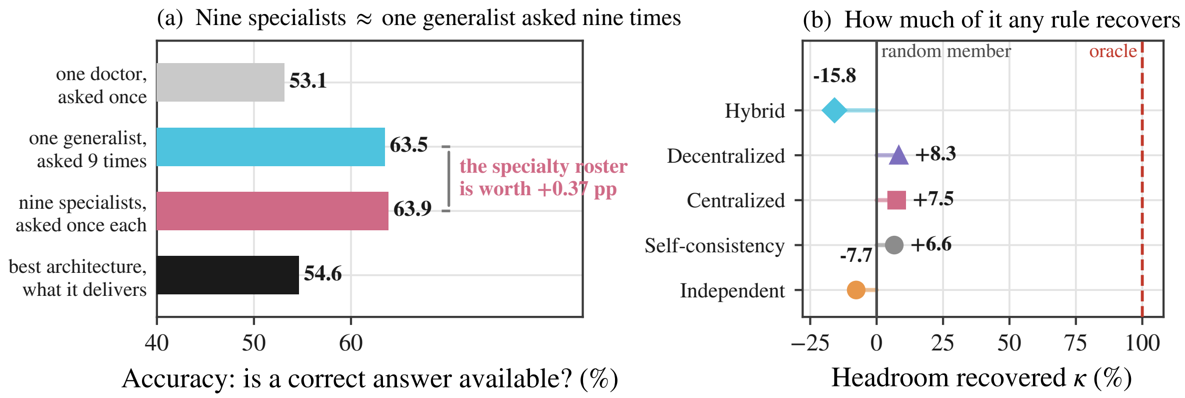

Figure 5: **A panel holds answers it cannot deliver.** (a) How often a correct answer is available at all, at n_a=9. Asking one generalist nine times makes one available 63.5% of the time; nine distinct specialists make one available 63.9%. The two middle bars are the comparison the design pattern rests on, and they are the same length: of the 9.3 pp of headroom a specialty panel appears to offer over a single attempt, 97% is sampling and 3% is the roster. The bottom bar is what the best architecture actually returns, 1.8 pp above one doctor. (b) The same gap on a selection scale, where 0 is picking one panel member at random and 100 is an oracle that always picks a correct one. Averaged over all configurations at $n_a \geq 3$, no rule reaches 9, and two fall below zero.

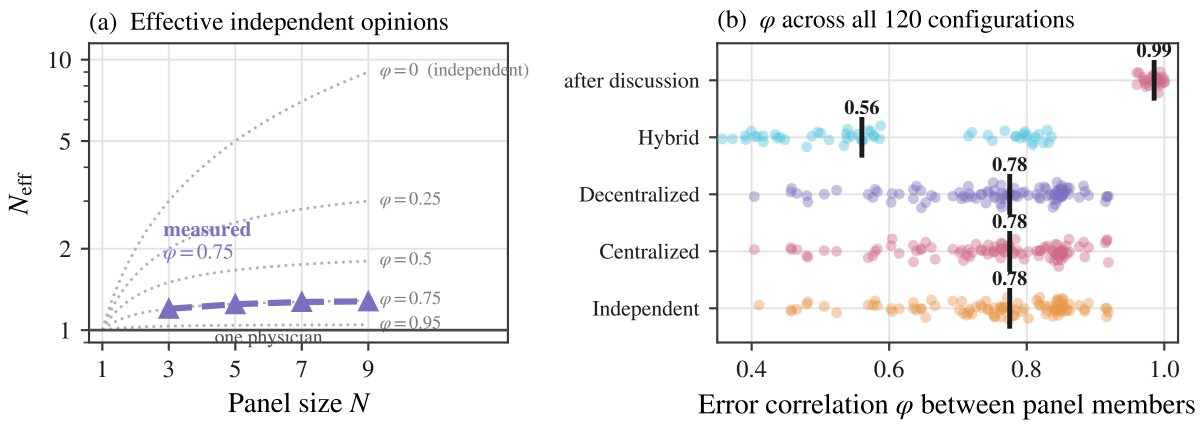

Figure 6: **Adding agents adds almost no information, and discussion removes what is left.** (a) Effective independent opinions $N_{\text{eff}} = N/(1 + (N-1)\varphi)$ against panel size, on a logarithmic scale. Dotted curves are contours of fixed error correlation φ; the solid line at N_{eff}=1 is a panel worth one physician. Measured panels track the φ≈0.75 contour, so nine members carry barely more than one opinion. (b) φ for every one of the 292 grid configurations, one point per configuration, with the median marked. The value is a near-constant of the setting rather than an artefact of any one cell, Hybrid routing is the only architecture that lowers it, and peer discussion drives it to 0.99.

racy, and it is worth them. The split is what identifies the mechanism. It tracks the communication channel: the roster carries 81% of the gain under independent voting, 40% under a central orchestrator and 22% under discussion. With no channel, distinct specialties are the only thing making two panelists disagree, and nine copies of one internist voting by majority are worth +0.50 pp—asking the same doctor again is not a second opinion. A communication channel manufactures the same disagreement the roster was manufacturing, and substitutes for it rather than adding to it. Either way, what the panel is buying is disagreement, not knowledge.

Putting the right specialist on the case does not help. The arithmetic above says a specialty roster buys little diversity. Medicine offers a more direct test of the same premise. Every MedXpertQA item is labelled with the body system it concerns, and the router ranks specialties by relevance, so at a given panel size some items receive a panel containing the specialty that owns their body system—a neurologist on a nervous-system case—

and others do not. If convening the right specialist is what a multidisciplinary panel is for, the first group should gain more.

We compare the *gain over the single doctor* on the same items, which absorbs item difficulty into the baseline. Across three models at $n_a \in \{3, 5\}$ the difference between panels that contain the matching specialty and panels that do not is $\mathbf{-0.05}$ pp on average, and one of the six contrasts reaches $p < 0.05$, which is what six comparisons produce by chance. A cardiologist on a cardiac case is worth nothing measurable, because the specialty prompt changes what the agent says it is and not what the model knows.

Half of the reported decorrelation is the ceiling, not the diversity. Any analysis that reads raw agreement or overlap as evidence of panel diversity is partly reading $\varphi_{\max}$ (§3.1), and a regression separates the two. Regressing the 45 model pairs in Table 4 on family identity and accuracy gap separates the two. On raw φ the accuracy gap dominates ($\hat{\beta} = -0.601$, 95% CI $[-1.034, -0.123]$, $R^2 = 0.341$). On $\varphi/\varphi_{\max}$ that coefficient vanishes and changes sign ($+0.156$, 95% CI $[-0.599, +0.994]$, $R^2 = 0.063$), while the family term is unmoved ($+0.093 \to +0.094$, both intervals excluding zero). Mixing capability does not decorrelate errors at all; it only lowers the ceiling the correlation could have reached. Mixing model families decorrelates genuinely, and by a little.

Independence is for sale, and the price is more than it is worth. The constructions in Table 4 order cleanly, and the ordering is the practitioner's menu (Fig. 7). Specialty prompts leave $\varphi/\varphi_{\max}$ at 0.766. Swapping in genuinely different checkpoints from one family reaches 0.560, and crossing training families inside a single vendor reaches 0.465—a nine-member panel worth 2.6 opinions, not nine. Crossing the vendor boundary, which is where the remaining hope for independence usually sits, does not continue the trend: every cross-vendor pair we can form among six independently trained ecosystems—OpenAI, Google, Anthropic, DeepSeek, Alibaba and Zhipu, 300 pairs over the same 750 items—gives $\varphi/\varphi_{\max} = 0.504$ and $N_{\text{eff}} = 2.29$, statistically indistinguishable from staying inside one vendor and pointing the wrong way ($\hat{\beta}_{\text{same vendor}} = -0.032$, 95% CI $[-0.088, +0.024]$ on normalised correlation, with the family term at $+0.088$). The comparison is not confounded by capability: the ladder pairs models at matched I on both sides of the boundary—`deepseek-v4-pro` at 59.7 against `gpt-5-mini` at 59.2, Gemini 3.5 Flash Lite at 50.3 against `gpt-5-nano` at 50.8—and restricting to pairs whose error rates differ by less than 5 pp, where $\varphi_{\max}$ averages 0.9 and the mechanical artefact nearly vanishes, leaves cross-vendor pairs at 0.480 against 0.386 within a vendor (2.02 against 2.31 effective opinions), if anything *more* correlated. Two models trained on different continents, on different corpora, under different alignment regimes, fail on the same medical items as reliably as two checkpoints from one lab.

The only way to make a panel disagree is to seat someone who is worse. Every remedy so far has failed to decorrelate the panel. One lever does work, and it is worth looking at closely because it is the one a practitioner would reach for next. Across the 360 model pairs, the wider the gap between two members' accuracies, the less their errors correlate: $r = -0.550$ ($p = 8 \times 10^{-30}$). Pairs within 5 pp of each other sit at $\varphi = 0.438$, a nine-member panel of 2.07 effective opinions; pairs more than 30 pp apart sit at $\varphi = 0.156$ and 4.35 effective opinions. Independence appears to double.

It does not. Normalise by the ceiling of Eq. 2 and the entire effect disappears: $r = -0.043$ ($p = 0.42$), with the two extremes at $\varphi/\varphi_{\max} = 0.487$ and 0.378. Two members with different error rates *cannot* correlate perfectly, so a capability gap lowers measured agreement whether or not it changes how either member reasons. What looks like diversity is arithmetic.

The deployment consequence is sharper than the statistical one. A panel of one strong, one mid and one weak specialist reaches the highest N_{eff} we measure and scores 40.8% under Centralized and 44.0% under Decentralized on MedXpertQA, against 49.2% and 47.6% for the same architectures with three strong members (§H.1). It disagrees more and answers worse. Disagreement produced by seating a weaker member is not a second opinion; it is a worse first one, and pooling cannot recover what was never there.

The headroom is item-level routing value, not a better specialty. Two oracles separate what a fixed roster could deliver from what per-item routing could. Always trusting whichever specialty slot turns out best over the whole benchmark—

Table 4: **Four sources of panel diversity.** Normalised correlation $\varphi/\varphi_{\max}$ is the quantity that is not confounded by unequal error rates. Rows relax, in turn, what makes two panel members different: a prompt, a checkpoint, a training family (`gpt-4o` / `gpt-4.1` / `gpt-5`), and finally the vendor—all pairs across six independently trained ecosystems, three American and three Chinese. The last step—the one usually assumed to be the answer—moves nothing, and moves it the wrong way. n is model–benchmark pairs.

Source of diversity	n	φ	$\varphi_{\max}$	$\varphi/\varphi_{\max}$	N^9_{eff}
One model, different specialty prompts	—	0.753	0.958	0.786	1.28
Different checkpoints, one training family	21	0.479	0.854	0.561	1.86
Different families, one vendor	39	0.308	0.663	0.465	2.60
Different *vendors* (OpenAI/Google/Anthropic/DeepSeek/Alibaba/Zhipu)	300	0.366	0.726	**0.504**	**2.29**
fully independent	—	0	1	0	9.00

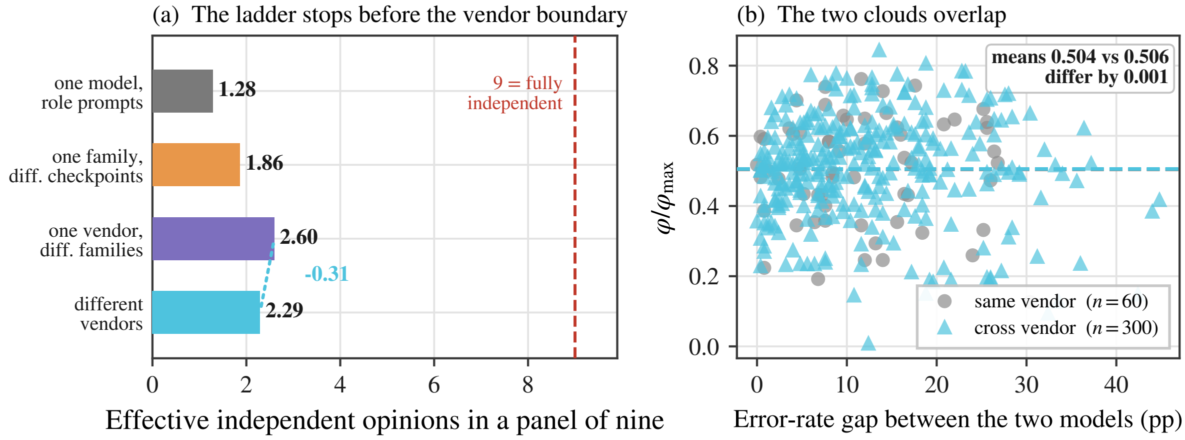

Figure 7: **Independence does not arrive at the vendor boundary.** (a) Effective independent opinions in a nine-member panel under four sources of diversity. Relaxing the prompt, then the checkpoint, then the training family lifts the panel from 1.31 to 2.60; relaxing the vendor moves it -0.31. (b) The evidence behind that mean: all 360 model–benchmark pairs, normalised correlation against the gap in the two members' error rates. The same-vendor and cross-vendor clouds occupy the same region and their means differ by 0.001. The rightward drift of both is Eq. 2 at work, which is why the normalised quantity is the one plotted.

a post-hoc choice no deployment could make in advance—is worth 54.4% against 53.1% for the single doctor. Choosing a correct opinion item by item is worth 63.9%. Of the 10.7 pp of headroom, 1.3 is available from picking a better specialist and 9.4 requires changing which member you believe from case to case. The design question is therefore not which specialists to convene but whether anything can arbitrate between them once convened.

4.4 Can the room tell who? The aggregation ceiling

The headroom is real. What reaches the output is not. Averaged over all configurations at $n_a \geq 3$, peer debate recovers 8.3% of it, an auditing orchestrator 7.5% and budget-matched self-consistency 6.6%, while majority vote recovers -7.7% and tiered referral $-\mathbf{15.8}\%$: the two commonest rules finish *below* the accuracy of a panel member drawn at random (Fig. 5b). Choosing the best architecture for each cell after the fact, which no deployment can do, reaches 17.5%. At $n_a{=}9$ the single doctor scores 53.1%, the oracle 63.9%, and the best measured architecture 54.9%. The rest of this section asks why.

Panels disagree at a human rate, and the disagreement is sterile. Two specialists in the same panel give different answers on 17.3% of items before any exchange (MedXpertQA 24.1%, MedAgentsBench 19.3%, MedQA 7.4%). That is close to the 21% of second opinions that produce a distinctly different diagnosis in a large second-opinion audit (Van Such et al., 2017), so on this measure an AI panel looks about as diverse as a pair of physicians. Resolving the pair against ground truth shows the resemblance is superficial. Across all pairs, 48.5% agree and are right, $\mathbf{38.7}\%$ agree and are *both wrong*, 9.6% split with exactly one right, and 7.7% split with both wrong. Of every disagreement, 44.5% is two agents reaching different wrong answers. The panel argues at roughly the human rate, but on two-fifths of items it has already agreed on the wrong diagnosis before the argument starts, and no aggregation rule and no panel size can reach those items.

Panels drop the answer they are holding. Error amplification A_e locates the failure inside the episode. Every architecture discards a substantial share—Centralized 0.29, Independent 0.27, Decentralized 0.23, Hybrid 0.08—and the ordering follows the capture rates rather than the panel sizes. Hybrid is lowest because it usually declines to convene a panel at all, so it has little to discard.

Redundancy says what the panel is doing instead. Mean rationale overlap among first opinions is 0.392 for pairs that are both right, 0.351 for pairs that are both wrong and 0.206 for pairs that split ($t = 119.5$ between the first two, $p < 10^{-300}$). Overlap separates *agreeing* from *disagreeing* by 0.15–0.19, and agreeing-and-right from agreeing-and-wrong by 0.040. It detects whether two specialists took the same path, and barely whether the path was correct. A panel that agrees is a panel whose members reasoned alike, which is the same fact as $\varphi = 0.76$ seen in the rationales rather than in the answers.

No signal the panel can compute tells it when it is right. Stated confidence is the obvious arbiter, and the natural objection is that a better one exists. We tested every quantity a panel can compute from its own first round: mean stated confidence, the entropy of the answer distribution, the majority share, the lowest member confidence and the spread of confidences. Against the outcome the arbiter needs to predict—whether this panel will answer correctly—all five land between AUC $=$ 0.570 and 0.604 (Table 5), a range of 0.034. The lowest member's confidence is marginally the best and still separates right from wrong with Youden $J = 0.179$. The problem is not the choice of signal. A panel whose members share 76% of their errors cannot distinguish the case where it agrees because the answer is easy from the case where it agrees because its members share a blind spot, and every statistic computable from its own outputs inherits that ambiguity.

Table 5: **Every arbiter a panel could use.** Discrimination of each first-round signal against whether the panel's final answer is correct, over all 64,591 panel episodes at $n_a \geq 3$. The spread across signals is 0.034 of AUC.

First-round signal	AUC	Youden J
Lowest member confidence	**0.604**	**0.179**
Mean stated confidence	0.603	0.166
Answer entropy	0.591	0.177
Majority share	0.590	0.177
Confidence spread	0.570	0.140

A selector trained on the answers does not beat counting votes. Nothing that reads the panel's output does better, including a model fitted on the answers. For each of the 64,591 panel episodes we formed one row per distinct candidate answer, with fifteen features computable from the panel's own first round: vote share, margin over the runner-up, the mean, maximum, minimum and spread of the confidences attached to that candidate, that candidate's confidence relative to the panel mean, rationale lengths, the number of distinct answers, the entropy of the vote distribution, and panel size. The label is whether the candidate is correct. We trained a logistic regression and a gradient-boosting classifier under five-fold cross-validation grouped by item: fitted on ground truth, tested in the same distribution, with access to everything the panel emits.

Neither beats majority vote. Selecting the highest-scoring candidate gives 52.46% (logistic) and 52.16% (boosted trees) against 52.48% for taking the plurality, while the oracle reaches 59.88%. On the capture scale the two learned selectors score -0.3 and -4.3: the extra features are noise. The 7.40 pp between plurality and oracle is not merely unexploited by the rules practitioners use; it is unexploited by a model fitted on the answers. Whatever distinguishes the panel's correct minority from its confident majority is not present in what the panel emits, which is why the productive direction is verification from outside the panel rather than a better rule for reading it.

The one architecture built to select cannot. Eq. 3 says the lever that matters is κ, and κ requires deciding which opinion to trust. Hybrid is the only architecture here that tries: a triage gate escalates to a panel when the first opinion's stated confidence falls below θ. It escalates on $\mathbf{21}\%$ of items overall, and the rate is set almost entirely by which model is behind the gate rather than by which items are hard—76.8% for `claude-sonnet-5` on MedXpertQA, 0.0% for `gpt-4.1-nano` on MedQA, where the gate never

Table 6: **Clinical safety.** Unanimity is measured on the last round with two or more agents; $P(\text{wr.}\,|\,\text{un.})$ is the chance a fully agreeing panel is wrong; the last column is how well stated confidence separates the panel's right answers from its wrong ones. The first row is the single doctor, for reference. Peer discussion buys near-total agreement and changes nothing about what that agreement is worth.

Architecture	unan.	$P(\text{wr.}\,|\,\text{un.})$	conf. AUC
Single doctor	—	—	0.599
Independent	67.5%	38.1%	0.607
Centralized	67.0%	38.1%	0.622
Decentralized	**98.0%**	44.9%	0.601
Hybrid†	51.2%	49.4%	0.598

the shaded row agrees on almost every item and is wrong on 45% of them. †Hybrid is measured on the 21% of items its triage gate escalates.

once fires in 1,000 items. This is not a poorly tuned threshold. It is the same measurement as §4.5 seen from the other side: a gate can only route on a signal that separates right from wrong, and stated confidence separates them with Youden $J = 0.166$. Hybrid is cheap and efficient because it is a single model most of the time, and it is a single model most of the time because there is no signal on which it could decide to be anything else.

4.5 Clinical safety: what consultation does to confidence

Accuracy is not the only thing coordination changes. In medicine the more consequential question is what it does to the *signals a clinician would act on*. Three measurements, all pointing the same way (Table 6).

Self-reported confidence does not separate right answers from wrong ones. A single doctor states 89.5 on the answers it gets right and 86.1 on the ones it gets wrong, which is AUC $= 0.604$ (Fig. 8). No architecture reads better: the five span 0.598 to 0.622, and no panel size or coordination topology moves them out of that band. The best achievable referral gate on stated confidence, calibrated before the main experiments, reaches a Youden J of 0.166. A clinician reading a panel's confidence learns almost nothing about whether to believe it.

Discussion manufactures consensus without manufacturing correctness. Unanimity, defined as every panelist emitting the same valid option in the last round with two or more agents, rises from 67.5% under independent voting to 98.0% under peer discussion at $N{=}9$. Conditional accuracy does not follow: $P(\text{wrong} \mid \text{unanimous})$ is 38.2% for Independent and 45.1% for Decentralized. A unanimous medical panel is wrong on 38–45% of what it agrees on, whichever architecture produced the agreement. Discussion does not change what unanimity is worth; it manufactures far more of it.

The most trustworthy-looking output is the least trustworthy. Taken together: consultation drives panels to near-total agreement, leaves the error rate given agreement at 45%, and never makes stated confidence a usable guide to whether the panel is right. The configuration a clinician would most readily act on, a unanimous and high-confidence multidisciplinary recommendation, carries essentially the same error rate as a divided one, and advertises itself more strongly. Accuracy gains of 6–8 pp on the best cells are real, and they arrive attached to a signal that cannot tell the clinician which of the remaining errors to look for.

4.6 What a panel costs

The architecture that wins on accuracy is the one that costs most. Successes per 1,000 tokens: Hybrid 0.81, Independent 0.41, Decentralized 0.37, self-consistency 0.27, Centralized 0.22. Centralized is 3.7× less token-efficient than the cheapest configuration, and the architecture responsible for every significant gain is the one that costs most per correct answer. Hybrid is the most efficient, because its triage gate declines to escalate most cases and therefore rarely pays for a panel at all.

On MedXpertQA at `gpt-5-nano`, Decentralized at $N{=}3$ reaches 38.0% while self-consistency at the closest matched dollar cost reaches 36.0%. Elsewhere the control wins or ties: on MedQA at the same tier, self-consistency at $k{=}15$ (88.4%) is within noise of the best panel (89.6%) at lower cost, and on MedAgentsBench at T1 self-consistency (20.0%) beats every architecture. Where coordination buys anything beyond repetition, it does so narrowly.

What a point of accuracy costs. Expressing each cell's best configuration as accuracy bought per dollar of additional spend makes the deployment rule concrete (Table 7, Fig. 12). The premium ranges from 1.9× to 11.5× the single doctor, and the return on it varies by two orders of

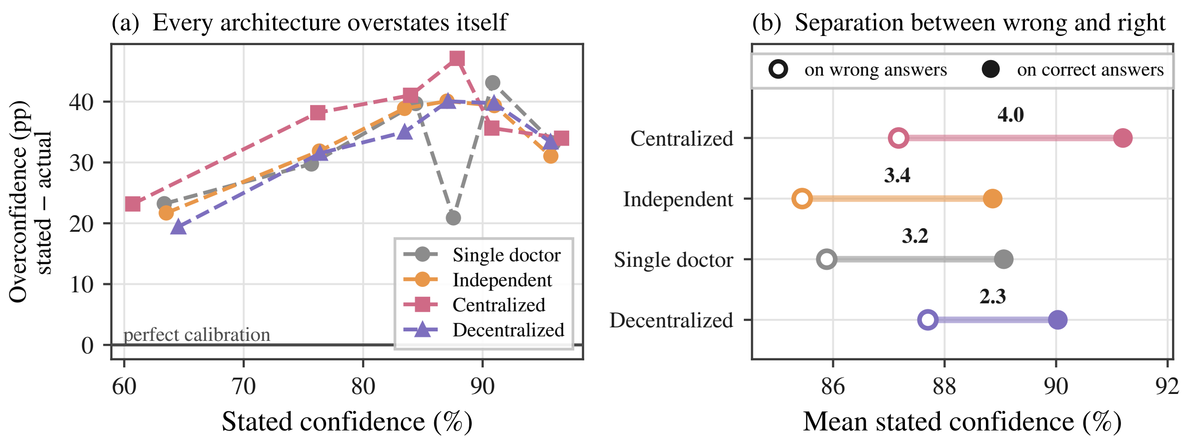

Figure 8: **Consultation degrades the signal a clinician would use to catch its errors.** (a) Overconfidence, defined as stated confidence minus the accuracy actually achieved at that confidence, against stated confidence. Perfect calibration is the solid line at zero; every architecture sits 19–47 pp above it at every confidence level, so panels are not merely uninformative about their own reliability but systematically overconfident. (b) The confidence gap between correct and incorrect answers at $N{=}9$, which is the quantity a reader would use to discount an answer. Peer discussion cuts it by a quarter, to below the single doctor's.

Table 7: **Accuracy per dollar.** For each cell, the best configuration of any architecture and panel size: its gain over the single doctor, the additional spend per 1,000 questions, the cost multiple, and the ratio.

Benchmark	Tier	gain	extra \$/1k	×cost	pp/\$
MedXpertQA	T1	+3.6	0.08	1.9	**44.5**
MedXpertQA	T2	+6.0	0.70	4.4	8.5
MedXpertQA	T3	+7.6	1.76	3.5	4.3
MedAgentsBench	T2	+2.4	0.28	2.5	8.5
MedAgentsBench	T3	+2.4	7.12	11.5	0.34
MedAgentsBench	T1	+1.2	0.39	6.2	3.1
MedQA	T1	+1.6	0.27	4.2	6.0
MedQA	T2	+1.6	0.49	4.1	3.3
MedQA	T3	+2.0	−0.14	0.8	—

magnitude: 8.5 and 4.3 percentage points per additional dollar per thousand questions on the two MedXpertQA cells with the largest gains, against 0.34 on MedAgentsBench at T3, where a +2.4 pp gain costs 11.5×. On MedQA at T3 the best configuration is Hybrid at $n_a{=}1$, which is *cheaper* than the single-doctor baseline because its gate almost never escalates.

5 Conclusion

A consultation is worth convening only if somebody in the room is right when the others are wrong, and the room can tell who. Our measurements say that panels of language models have a little of the first and almost none of the second.

The first is scarcer than the design pattern assumes, and it does not come from where the design pattern puts it. Agents prompted into different specialties make the same mistakes ($\varphi = 0.758$), leaving a nine-member panel with 1.27 effective independent opinions, and peer discussion pushes them into near-perfect agreement ($\varphi = 0.983$), leaving a nine-member consultation with 1.01. Of the 10.7 pp that separates one doctor from the best answer anywhere in a nine-member panel, 97% is already available from asking one generalist the same question nine times; the specialty roster—the defining feature of the systems built on this pattern—contributes +0.37 pp. Independence can be bought, but not cheaply and not usefully: three independently trained model families reach 2.60 effective opinions out of nine and swapping vendor entirely—300 pairs across six ecosystems on two continents—reaches 2.29, and the least correlated panel we can assemble reaches that state by seating a much weaker member and then performs like the stronger member working alone.

The second is the binding constraint. On a scale where picking a panel member at random scores 0 and always picking a correct one scores 100, an auditing orchestrator scores 4.2, peer debate 8.3, and majority voting and tiered referral score below zero, at −7.7 and −15.8. The panel usually holds the answer it does not give. The quantity such a system would need—a signal separating a right member from a wrong one—is missing: self-reported confidence separates them with Youden $J = 0.166$, and the one architecture here that tries to route on it convenes a panel on 21% of items and never once on one benchmark–model cell. Collaboration gain factors as available signal times the fraction recovered, and both factors are small.

Three knobs the field is turning, and one it is not. Panel size: over $N \in \{3, \ldots, 9\}$ the effective number of opinions moves from 1.16 to 1.22, and panel size alone predicts gain at cross-validated $R^2 = -0.006$. The specialty roster: a temperature setting reproduces 99% of the diversity it buys, and with the roster removed entirely a panel of identical internists still collects 60% of the accuracy the specialist panel delivers. Coordination topology: pairwise architecture contrasts are non-significant, so star, complete, tiered and edgeless are indistinguishable from one another. The knob that is not being turned is verification: every point of headroom the field is trying to reach through richer coordination protocols is already present in the panel and is being discarded for want of a way to recognise it. Where a panel is deployed anyway, the decision should be made by measuring the single-model baseline on the intended case mix first—budget-matched self-consistency matches or beats every architecture we tested at a fraction of the cost—and capability belongs on the agents doing the reasoning rather than the one supervising them, which matters $4.1\times$ more.

The broader caution is about what a consulting system advertises. A unanimous, high-confidence multidisciplinary recommendation is the output a clinician is most likely to act on, and in these experiments it is the output whose reliability is least legible from its own signals: unanimity reaches 98% while the error rate given unanimity stays at 45%, and stated confidence separates the panel's right answers from its wrong ones at AUC ≤ 0.62 under every architecture we tested. Making medical panels agree is easy. Making them *decidable* is the open problem.

6 Limitations and future works

Our tasks are multiple-choice examination items, not clinical encounters. They carry no history taking, no test ordering, no longitudinal follow-up, and the exact-match endpoint rewards the option-selection component of diagnosis only. The $T = 0$ property that makes the design clean also bounds what it licenses: results here do not extrapolate to tool-using clinical agents. Message density explains almost none of the variance in performance ($R^2 = 0.056$) in this format, and open-ended settings where agents must generate a differential and request evidence give coordination more to carry; we would expect those metrics to matter more there. The selection result does not depend on the format—whatever a panel produces, something has to decide which member to believe.

The diversity ladder spans six independently trained ecosystems—OpenAI, Google, Anthropic, DeepSeek, Alibaba and Zhipu—and measures the vendor boundary rather than assuming anything about it (§4.3); it does not span all of them. Two readings remain open. A vendor whose training corpus or objective departs further from the current consensus than Gemini does from GPT might decorrelate where these two do not, and our grid cannot exclude it. And our benchmarks are static text items, so a vendor difference that showed up only in tool use, retrieval or hedging behaviour would not register here. What the measurement does close is the specific hope that motivated the question—that pooling across the major commercial labs is what makes a panel independent. Between the two largest, at matched capability, it is not.

Panels are nested by construction (§3.2), so the reported curves estimate accuracy as a routed panel is extended, not the expected accuracy of independently redrawn panels at each size. The coupling reduces variance on the architecture contrasts we care about and does not bias within-item paired tests, but a reader interested in the independent-panel estimand should read the curves accordingly.

The specialty router ranks by relevance, so larger panels append progressively less relevant specialists, which entangles "more experts" with "weaker experts" in the specialty-role condition. The generic-internist control separates them (§4.3): with the roster removed and panel size the only variable, the panel still gains $+2.77$ pp, and the roster's own contribution, $+1.87$ pp, is largest exactly where no communication channel exists. The entanglement is therefore measured rather than assumed, and it does not run in the direction that would manufacture our result—a roster of *decreasing* relevance still adds accuracy.

Difficulty strata are defined by the pass rate of mid-tier models, some of which also appear in the evaluation ladder. This inflates the level of the hard-stratum results but not the within-item slope across panel sizes, which is computed on identical items.

Ethics Statement

This work uses only public benchmark items derived from licensing examinations and published case collections. No patient data, no human subjects and no clinical deployment are involved.

The findings carry a deployment risk that we want to state plainly rather than leave implicit. We show that multi-agent consultation systems manufacture the *appearance* of corroboration without the substance: peer discussion drives unanimity from 66% to 98% while the error rate given unanimity stays at 45%, and no signal the panel emits separates its right answers from its wrong ones at better than AUC $= 0.61$. A unanimous, high-confidence multidisciplinary recommendation is simultaneously the output a clinician is most likely to act on and, among the outputs we measure, the one whose reliability is least legible from its own signals. Anyone deploying such a system owes its users an explicit statement that panel agreement is not evidence of correctness, and should not surface consensus or confidence as a trust cue in a clinical interface. The same property makes these systems attractive for producing medical misinformation that reads as authoritative, since apparent expert consensus is cheap to manufacture and, as we show, uninformative.

Our measurements point to three specific mitigations rather than a general caution. First, do not surface unanimity or aggregate confidence as a trust cue: in our data a unanimous panel is wrong 50% of the time, and no statistic a panel can compute about itself—stated confidence, answer entropy, majority share, minimum member confidence—separates its right answers from its wrong ones better than AUC $= 0.60$. An interface that reports "all five specialists agree" is reporting a quantity with almost no diagnostic content while implying the opposite. Second, surface the panel's *disagreements* instead, and surface them unresolved: on 9.6% of items exactly one member is right, which is where the panel's whole value lies, and a resolved consensus destroys precisely that information before the clinician sees it. Third, route verification outward. The one intervention our diversity ladder shows to be large is a judgement from outside the model ecosystem, consistent with the hybrid human–model collectives of Zöller et al. (2025); verification against retrieved evidence or a clinician is a different kind of signal from more agreement among models, and only the different kind changes the arithmetic.

We report the failure mode as a measured property of systems already being built and deployed, not as a novel capability.

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A Model capability index

Prior work scores models on a published Intelligence Index and reports that a *task-grounded* capability metric explains more variance ($R^2 = 0.413$) than the generic one ($R^2 = 0.373$). We therefore use the task-grounded form throughout: I is a model's mean single-doctor chain-of-thought accuracy across the three benchmarks. Table 8 gives the components. The three tiers span $I = 34.0$ to 59.2, and their single-benchmark baselines span 15.6% to 91.6%, crossing the 45% region from both sides. That crossing is what makes the capability-ceiling test possible at all.

B Domain complexity

B.1 Complexity metric construction

Domain complexity $D \in [0, 1]$ is the arithmetic mean of three complementary measures (Kim et al., 2026): the **performance ceiling** $1 - p_{\max}$, where $p_{\max}$ is the highest performance any evaluated system reaches; the **coefficient of variation** σ/μ of performance across all configurations, a scale-invariant measure of relative spread; and the **best-model baseline** $1 - p_{\text{best}}$, where p_{best} is the strongest single-model performance on the dataset.

B.2 Domain characterisation

Table 9 gives the components. MedQA is a low-complexity domain ($D = 0.097$); MedXpertQA and MedAgentsBench-hard are high-complexity and nearly indistinguishable ($D = 0.489$ and 0.478).

B.3 The critical threshold does not transfer

A critical complexity threshold at $D \approx 0.40$ has been reported for general-domain agents (Kim et al., 2026): below it multi-agent architectures yield net positive returns, above it coordination overhead consumes resources otherwise spent on reasoning. Our data do not support a benchmark-level threshold of any value. MedXpertQA ($D = 0.489$) and MedAgentsBench-hard ($D = 0.478$) differ by 0.011 in complexity yet by 3.5 pp in mean multi-agent gain ($+3.09$ pp versus -0.37 pp), giving opposite signs at effectively the same D. A benchmark-level statistic cannot separate them because the quantity that does separate them varies *within* each benchmark: on MedAgentsBench-hard the per-tier gains run -2.78, $+1.02$, $+0.66$ pp as P_{SA} moves from 20.4% through 32.4% to 44.4%. Domain complexity is the right idea at the wrong granularity for medicine. The configuration-level baseline P_{SA} is what carries the signal.

C Datasets

MedXpertQA-Text (Zuo et al., 2025): 2,450 expert-level items, ten options each, tagged by 12 body systems and 3 task types. We draw a stratified sample interleaved across (body system $\times$ task) so that any prefix is approximately balanced, then take the first 250. **MedQA-USMLE** (Jin et al., 2021): 1,273 four-option test items; randomly shuffled with a fixed seed, first 250. **MedAgentsBench-hard** (Tang et al., 2025): the contamination-screened hard split, 894 items after excluding the `medqa_5options` subset, stratified-interleaved across the ten source subsets, first 250. All samples are nested prefixes, so a larger sample re-uses every call of a smaller one.

Item difficulty is additionally tagged easy/medium/hard by the pass rate of three mid-tier models at k=5; on the MedXpertQA sample this yields 13/62/125 items.

D Implementation details

D.1 Technical infrastructure

All models are accessed through one OpenAI-compatible client with a content-addressed disk cache, exponential backoff with rate-limit-aware retry, a per-call JSONL ledger recording tokens, latency and cost, and a cross-process spend journal guarded by an OS file lock. Every LLM call is cached on the full request (model, system, messages, temperature, seed, max tokens, effort, JSON mode, tag), so a re-run is a replay.

Table 8: Capability index and its components (single-doctor CoT accuracy, %).

Model	MedXpertQA	MedAgentsBench	MedQA	I (mean)	Setting
OpenAI — the capability ladder the grid is run on					
`gpt-4.1-nano`	15.6	20.4	66.0	**34.0**	—
`gpt-5-nano`	32.0	32.4	88.0	**50.8**	effort=low
`gpt-5-mini`	41.6	44.4	91.6	**59.2**	effort=low
Google — a second vendor, for the diversity ladder					
`gemini-3.5-flash-lite`	34.8	31.2	84.8	**50.3**	—
`gemini-3.7-flash`	60.4	56.4	95.8	**70.9**	—
Anthropic — a third vendor, for the diversity ladder					
`claude-haiku-4.5`	23.2	20.8	79.6	**41.2**	—
`claude-sonnet-5`	49.6	47.6	94.0	**63.7**	—
DeepSeek — a fourth vendor in the grid; Alibaba and Zhipu contribute single-doctor baselines to the diversity ladder					
`deepseek-v4-flash`	37.2	39.6	88.0	**54.9**	—
`deepseek-v4-pro`	44.8	42.0	92.4	**59.7**	—
`qwen3.8-flash`	34.4	32.8	88.8	**52.0**	—
`qwen3.8-max`	42.0	40.0	88.8	**56.9**	—
`glm-5.3-flash`	47.6	45.6	93.2	**62.1**	—
`glm-5.3`	42.8	46.4	93.6	**60.9**	—

Capability-matched cross-vendor pairs:
`gemini-3.5-flash-lite` (50.3) with `gpt-5-nano` (50.8), and
`deepseek-v4-pro` (59.7) with `gpt-5-mini` (59.2) across the Pacific.

Table 9: Domain complexity of the three medical benchmarks.

Benchmark	$1 - p_{\max}$	σ/μ	$1 - p_{\text{best}}$	D	mean MAS gain
MedXpertQA	0.360	0.334	0.396	**0.363**	+1.36 pp
MedAgentsBench-hard	0.412	0.328	0.436	**0.392**	-0.34 pp
MedQA (USMLE)	0.036	0.108	0.036	**0.060**	+0.72 pp

Table 10: **Model selection probe.** The eight candidates screened on 40 items per benchmark with single-doctor CoT *before the grid was run*, which is how the capability ladder was chosen: three OpenAI checkpoints, monotone and straddling the 45% threshold, with two Google models added for the vendor contrast. The Anthropic and DeepSeek models entered later and were not screened this way; they went straight into the grid, and their measured accuracies are in Table 8 with everything else.

Vendor	Model	Setting	Role	MedXpertQA	MedQA	USD / 1k items
Selected						
OpenAI	`gpt-4.1-nano`	—	tier T1	25.0%	75.0%	0.09
OpenAI	`gpt-5-nano`	effort=low	tier T2	37.5%	95.0%	0.17
OpenAI	`gpt-5-mini`	effort=low	tier T3	55.0%	97.5%	0.64
Google	`gemini-3.5-flash-lite`	—	vendor contrast	30.0%	95.0%	0.09
Google	`gemini-3.7-flash`	—	vendor contrast	80.0%	97.5%	0.41
Screened, not selected — used only for the pairwise φ analysis						
OpenAI	`gpt-4o-mini`	—	—	20.0%	87.5%	0.13
OpenAI	`gpt-4.1-mini`	—	—	32.5%	85.0%	0.41
OpenAI	`gpt-5.4-nano`	effort=low	—	47.5%	95.0%	0.14

D.2 Agent configuration

Single agents answer in one call. Independent panels deploy N specialists with synthesis-only aggregation. Centralized systems use one orchestrator over N sub-agents with a maximum of 2 orchestration rounds and one re-task per round. Decentralized systems run N agents through up to 3 discussion rounds, stopping early only on true unanimity. Hybrid systems combine a generalist triage gate with the specialist panel and escalate to full discussion when the panel has no majority. Agent temperature is 0.7 within panels, 0.3 for single-doctor baselines and for orchestrator/moderator turns; reasoning models are called at `effort=low` with a minimum completion budget of 1,600 tokens, since reasoning tokens are drawn from the same budget and a smaller cap returns empty output.

D.3 Prompt compilation

Every agent receives the same rendered item (stem, then options as `(A) ...`) and the same answer schema: JSON with `answer`, `confidence` (0–100) and `reason` (at most 45 words). Only the system prompt differs, naming the specialty. Peer context in communicating architectures is capped at a fixed 900-character budget, balanced-truncated across peers and *independent of* N, so a panel-size effect cannot be a context-length effect. Peer ordering is deterministically rotated per (item, round) so that the first-listed speaker is decorrelated from the top-ranked specialty.

D.4 Evaluation methodology

Sample sizes. 250 items per benchmark, 480 multi-agent configurations, 161,999 episodes. **Controls.** All architectures share the item rendering, answer schema, role-prompt template and decoding parameters; only coordination logic differs. Model weights are frozen. The primary endpoint is exact match against the dataset's own gold option key. No LLM judge appears anywhere in the evaluation path. Episodes that fail for infrastructure reasons carry an error status and are excluded from accuracy rather than scored as wrong, which would bias call-heavy architectures downward. Cost accounting is nominal: cached calls are charged at list price, so reported economics describe what a configuration would cost to run.

E Complete results grid

The per-configuration grid — 420 multi-agent configurations and their single-model controls, with accuracy, confidence interval and cost for each — is released with the code rather than printed here, together with the raw episodes it is computed from.

F The published scaling form, fitted to medical data

The main text measures collaboration gain directly. For readers who want the comparison, we also fit the scaling form proposed for general-domain agent systems (Kim et al., 2026) to our grid. It is reported here rather than in the results because none of the paper's claims rest on it.

With accuracy as the response, the fit is degenerate. Fitting that form directly gives $R^2 = 0.997$ (5-fold cross-validated $R^2 = 0.996$), which is an artefact of our design rather than a result. The response is accuracy and one predictor is the single-doctor baseline P_{SA}; the same items and the same model underlie both terms and only the coordination structure varies, so P_{SA} alone nearly determines accuracy ($\hat{\beta}_{P_{\mathrm{SA}}} = 0.875$, $p < 10^{-4}$). We therefore refit with the collaboration *gain* as the response, which is what the scaling principle is actually about.

Coordination metrics carry the fit. With gain $= P - P_{\mathrm{SA}}$ as the response, $R^2 = 0.638$ and 5-fold cross-validated $R^2 = 0.415$. Coordination metrics carry that fit with no tool-count term at all. Coefficients ($n = 225$ configurations including the self-consistency control, benchmark-clustered SEs):

term	$\hat{\beta}$	p
$\log(1 + A_e)$	-0.150	< 0.001
R (redundancy)	$+0.082$	< 0.001
c (message density)	$+0.006$	< 0.001
$\log(1 + O\%)$	$+0.005$	0.024
$P_{\mathrm{SA}} \times \log(1{+}n_a)$	$+0.027$	0.048
$\log(1 + n_a)$	-0.003	0.701
P_{SA}	-0.113	0.051
P_{SA}^2	$+0.037$	0.390

Error amplification is the dominant penalty; redundancy and message density carry small positive effects; overhead is not penalised once $T{=}0$; and neither panel size nor P_{SA} has a main effect.

$T_{\text{tools}} = 0$ throughout, so the $\log(1 + T)$ main effect and the $O\% \times T$ and $E_c \times T$ interactions vanish identically. Coordination metrics remain highly significant without them, which establishes that the coordination effect is not a re-description of tool overhead. The coefficient on $\log(1 + O\%)$ turns *positive* here. With no tools to coordinate over, spending more tokens on deliberation is mildly beneficial rather than costly.

Robustness. Our fit carries I and I^2 with large coefficients of opposite sign (-0.231 and $+0.228$), but bootstrap resampling ($n = 1000$) gives them standard errors of 0.94 and 0.54. With three capability tiers the quadratic is unidentifiable and we do not interpret it. Every other large coefficient is stable: $\log(1 + A_e)$ has bootstrap SE 0.024 and P_{SA} has 0.044.

Residuals pass a normality test (Shapiro–Wilk $p = 0.207$) but fail homoscedasticity decisively

(Breusch–Pagan $p = 0.0001$). All standard errors in this paper are therefore heteroscedasticity-robust (HC3), clustered by benchmark where more than two clusters exist. Residual standard error is $\hat{\sigma} = 0.017$, reflecting that collaboration gains in medicine are small in absolute terms. Regularised alternatives do not improve prediction: Lasso (10-fold CV for λ) retains 9 of 11 predictors at CV $R^2 = 0.382$ and Ridge reaches 0.144, both below the unregularised 0.415, so we retain the full specification.

Nested model comparison isolates where the explanatory power lives (5-fold CV R^2): the full model reaches 0.415; capability alone 0.196; coordination metrics alone 0.093; the single-doctor baseline alone 0.043; and **panel size alone** -0.006, worse than predicting the mean. Any account of medical consultation that treats "how many agents" as the design variable is fitting noise.

G Coordination dynamics

Figure 9 plots all three quantities.

Coordination is cheap in turns and uninformative in messages. Fitting $T = a(n + 0.5)^b$ gives $a = 0.84$, $b = 0.777$, $R^2 = 0.327$. The exponent is sub-linear, so turns grow more slowly than agents. A clinical panel answering a closed question terminates on unanimity rather than negotiating open-ended tool use. Turn counts by architecture are Decentralized 6.8, Centralized 6.2, Independent 5.0, Hybrid 2.1, against a single-agent 1.4, a range of 1.5 to 4.9×. The hard resource ceiling they infer from super-linear growth does not arise in tool-free consultation.

$S = 0.504 - 0.028 \ln(c)$, $R^2 = 0.025$: message density explains almost none of the variance in accuracy. Our Centralized panels run at $c = 0.88$ and Decentralized at $c = 0.60$, well past the $c^* = 0.39$ plateau reported elsewhere, with no accuracy penalty. Once a panel is communicating at all, communicating more neither helps nor hurts.

Coordination failure needs a channel to occur on. We classify first-round errors into four categories, following the multi-agent failure taxonomy of Cemri et al. (2025):

	Indep.	Centr.	Decentr.	Hybrid
Logical contradiction	6.3%	6.3%	6.3%	3.4%
Numerical drift	4.5%	4.7%	4.8%	3.1%
Context omission	13.0%	13.3%	13.4%	6.8%
Coordination failure	**0.0%**	3.0%	1.2%	0.0%

Coordination failure is **0.0%** under Independent coordination, and necessarily so: with $C = \emptyset$ there is no channel on which it could occur. Hybrid is also at 0.0% because its triage gate declines to convene a panel on most items. The magnitudes do not replicate: peer verification is reported to cut logical contradiction from 16.8% to 11.5%, whereas our first-round contradiction rate is flat at 6.3% across communicating and non-communicating architectures. Numerical drift is an order of magnitude rarer than in tool-using tasks (3–5% against 19–26%), as expected when no arithmetic is cascaded.

H Additional results

Three views of the grid that the main text states in prose. Figure 10 gives the full panel-size curve behind the claim that gains saturate at $n_a = 3$; Figure 11 shows that the spread within an architecture dwarfs the difference between architectures, which is why the pairwise contrasts in §4.2 so rarely separate; and Figure 12 plots the cost–performance frontier summarised in Table 7.

H.1 Where capability belongs inside a panel

A separate arm varies the capability of the panel's members rather than its coordination structure (Fig. 13).

Specialist capability matters four times more than the orchestrator's. Replacing the whole panel with weak models costs 34.4 pp (49.2% to 14.8%). Splitting that substitution isolates where the loss comes from. Downgrading only the *orchestrator*, keeping strong specialists, costs 3.6 pp, so the panel retains 90% of the homogeneous-high span. Downgrading only the *specialists*, keeping a strong orchestrator, costs 14.8 pp, retaining 57%. A strong attending physician recovers more than half of what a weak panel loses, but a weak attending physician barely dents a strong one. The asymmetry is 4.1×. In a centralized clinical panel, capability should therefore be spent on the agents doing the reasoning rather than on the agent doing the auditing.

An orchestrator too weak to audit is worse than no orchestrator at all. The largest single degradation anywhere in this study is a Centralized panel with a weak model on MedQA at $P_{\text{SA}} = 66.0\%$: -13.60 pp, $p = 10^{-5}$. Inspection of those episodes shows the mechanism directly. The orchestrator overrides a correct specialist majority.

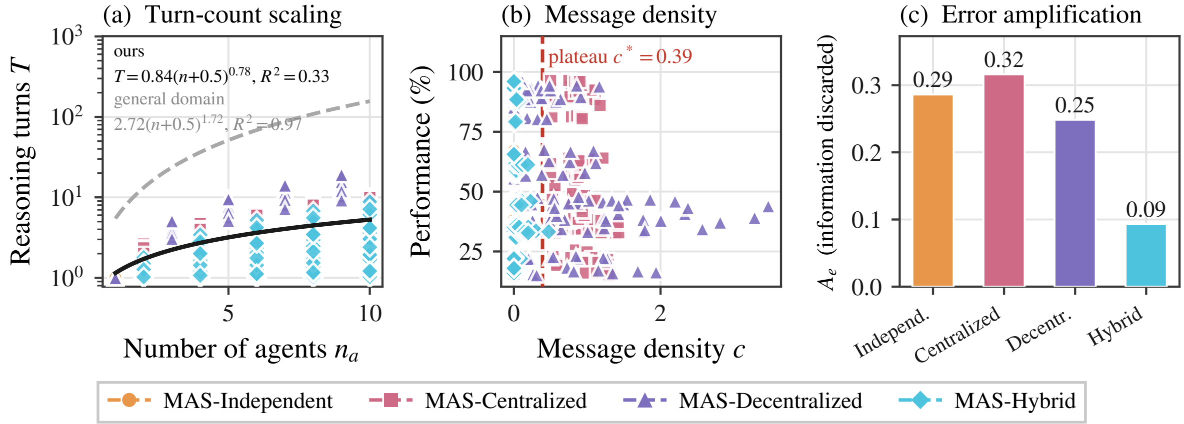

Figure 9: **A panel discards what it holds.** (c) Error amplification A_e: the share of correct first opinions a panel emits a wrong answer over, normalised by the single-agent error rate. (a) and (b) place the setting against the published general-domain law: medical panels converse an order of magnitude less than it predicts, and performance is uncorrelated with how much the panel says (§G).

This is the error-amplification term in its most extreme form. It is also the counterpart to the previous result. The audit is what makes centralized coordination work, so an auditor that cannot audit converts the architecture's strength into its worst liability.

Mixed-capability panels land between their components, with no emergent benefit. A panel of one strong, one mid and one weak specialist reaches 40.8% under Centralized and 44.0% under Decentralized, against 49.2% and 47.6% for homogeneous-high. Both sit below the strong homogeneous panel and well above the weak one, roughly where a capability-weighted average would put them. We find no medical analogue of the heterogeneous-mixing advantage reported for one model family (Kim et al., 2026), in which a low-capability orchestrator with high-capability sub-agents outperformed a homogeneous high-capability panel by 31%. In our data that configuration is the second-best Centralized cell but still 7% below homogeneous-high, directionally matching what they observe for OpenAI models, where heterogeneous centralized configurations degrade.

H.2 Where in medicine collaboration pays

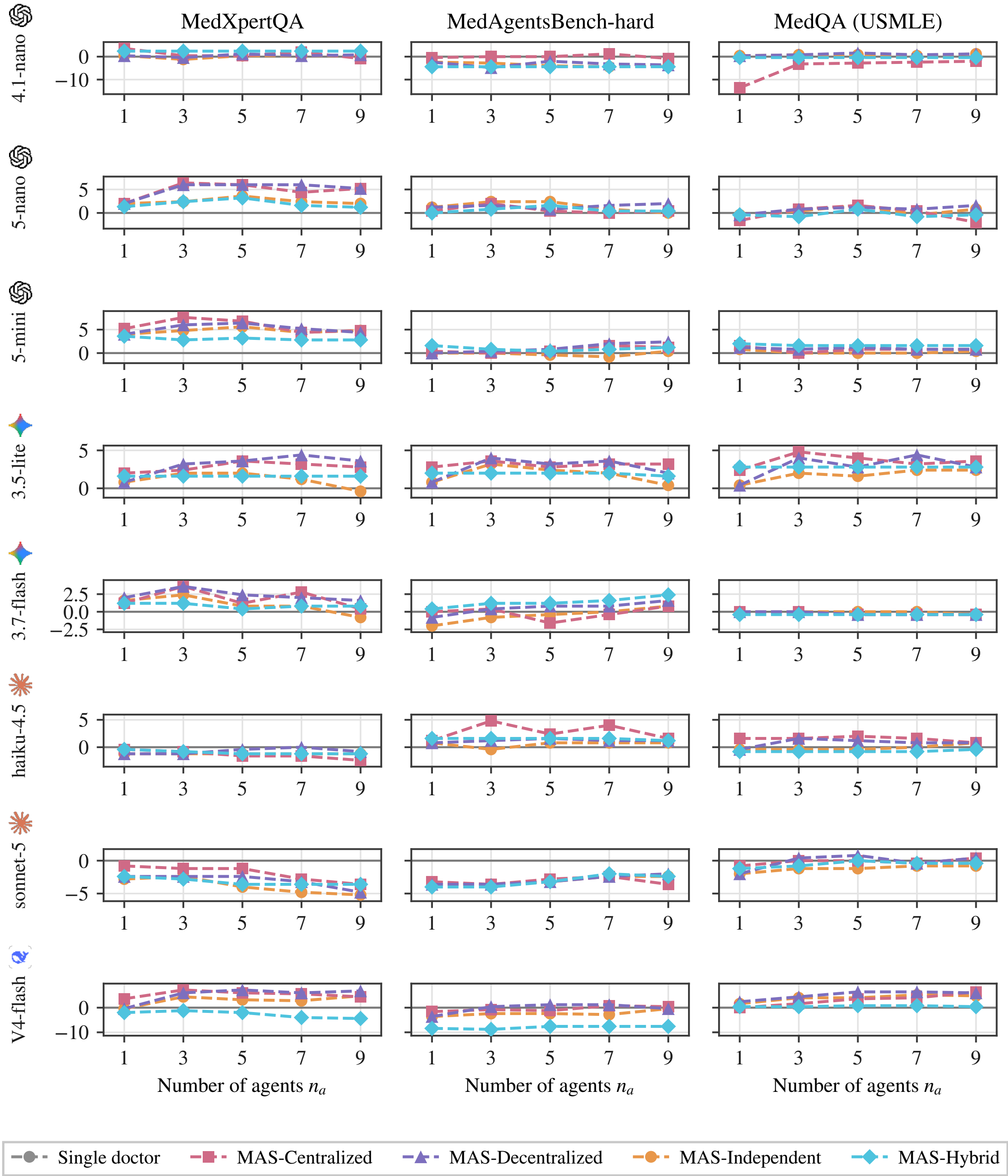

Figure 10: **Number-of-agents scaling.** Gain over the single doctor against panel size $n_a \in \{1, 3, 5, 7, 9\}$, for every architecture, model and benchmark; rows are models, columns benchmarks, and all panels share one vertical scale. Every series stays within a few points of the baseline across the whole range, and where gains occur they are captured by $n_a = 3$.

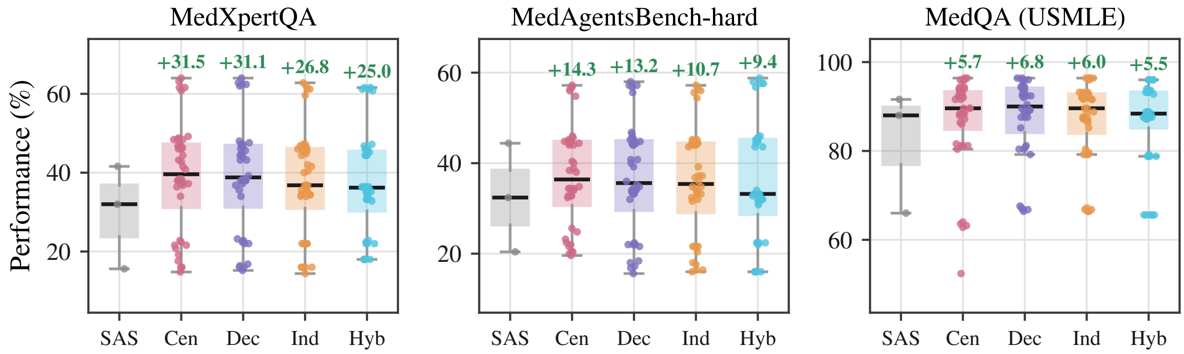

Figure 11: **Single doctor versus panel: performance distributions.** Box plots over all configurations of each architecture, with individual configurations overlaid. Bars are labelled SAS (single doctor), Cen, Dec, Ind and Hyb for Centralized, Decentralized, Independent and Hybrid. Annotations give the change in mean relative to the single doctor, as a percentage of that baseline. The spread within an architecture dwarfs the difference between architectures.

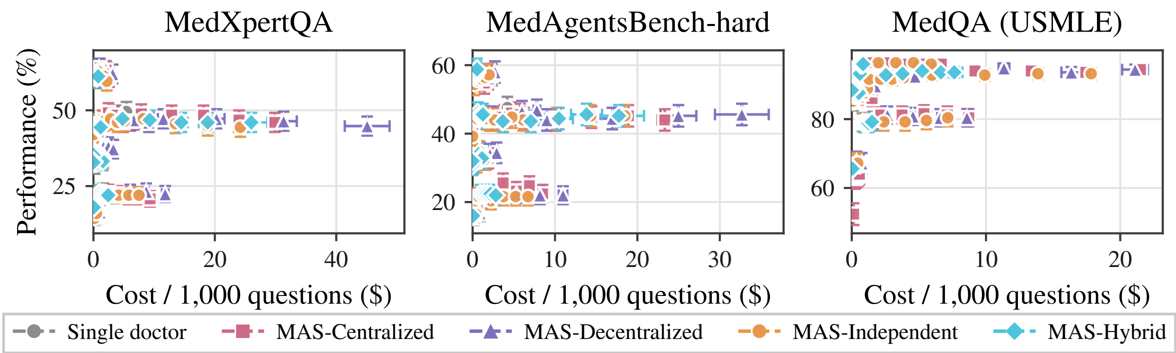

Figure 12: **Cost–performance trade-offs.** Mean performance against nominal cost per 1,000 questions, with SEM error bars on both axes. Each point is one configuration.

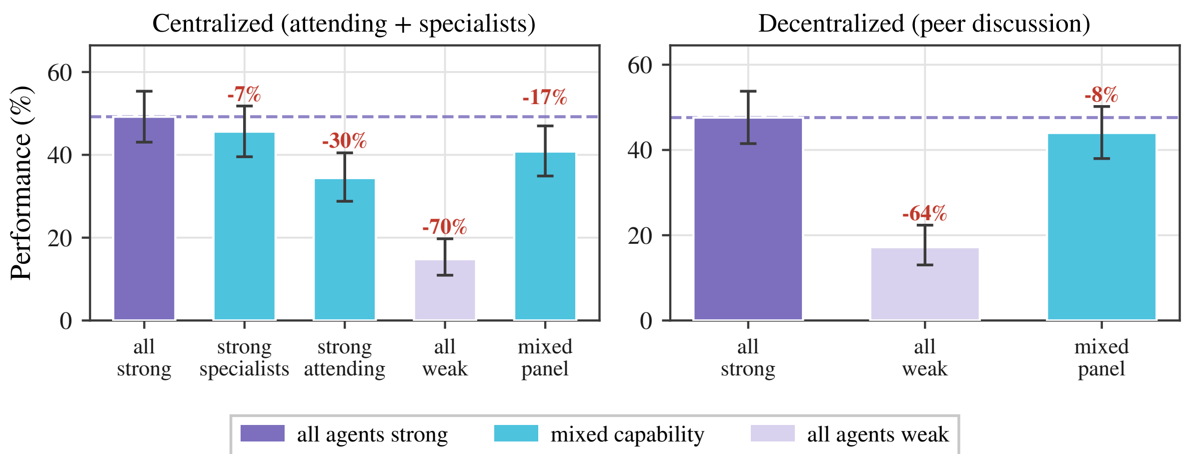

Figure 13: **Agent heterogeneity.** Performance of Centralized and Decentralized panels at n_a=3 on MedXpertQA under five capability placements. Strong is `gpt-5-mini`, weak is `gpt-4.1-nano`; *strong specialists* pairs a weak attending with strong specialists and *strong attending* reverses that pairing. Annotations give the change relative to the all-strong panel (dashed line). Decentralized coordination has no attending, so the two attending-capability conditions are structurally undefined there and are omitted.